



Experiment 7

Normalization · Candidate Keys · Decomposition

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01 AIM

Identify candidate key(s), prime/non-prime attributes and the highest normal form of a given relation and its FDs; decompose relations below 3NF/BCNF to eliminate redundancy.

02 TOOLS USED

- Manual FD-closure analysis (pen & paper)
- MySQL Workbench — optional schema verification

03 THEORY — QUICK REFERENCE

Core Definitions

FD ($X \rightarrow Y$): X uniquely determines Y — no two tuples agree on X but differ on Y.

Candidate Key: minimal attribute set whose closure = all attributes.

Prime / Non-prime: prime = part of some candidate key; everything else is non-prime.

Normal Form Conditions

NF	Condition
1NF	All attribute values atomic — no repeating groups.
2NF	1NF + no non-prime attribute has a partial dependency on a candidate key.
3NF	2NF + every non-trivial $X \rightarrow Y$ has X as superkey, or every Y-attribute is prime.
BCNF	3NF + every non-trivial $X \rightarrow Y$ has X as a superkey (no prime exception).

A relation is assigned the highest NF whose condition holds, given all lower ones hold too.

04 PRACTICAL SET A — CANDIDATE KEY & HIGHEST NF

Q1 R(A,B,C,D,E)

FDs: $CE \rightarrow D$ | $D \rightarrow B$ | $C \rightarrow A$

C, E never appear on a right-hand side $\rightarrow$ compulsory. $(CE)^+ = \{C, E, D, B, A\} = R$, so CE is the only key.

FD $C \rightarrow A$: C is a proper subset of key CE, and A is non-prime $\rightarrow$ partial dependency $\rightarrow$ 2NF fails.

Candidate Key	CE
Prime	C, E
Non-prime	A, B, D
Highest NF	1NF

Q2 R(A,B,C,D,E,F)

FDs: $AB \rightarrow C$ | $DC \rightarrow AE$ | $E \rightarrow F$

B, D never appear on a RHS $\rightarrow$ compulsory but insufficient alone.

$(ABD)^+$ = all attrs, and $(BCD)^+$ = all attrs $\rightarrow$ two minimal keys.

Subset CD ($\subset$ key BCD) has $(CD)^+ = \{A, C, D, E, F\}$, deriving non-prime E, F $\rightarrow$ partial dependency $\rightarrow$ 2NF fails.

Candidate Keys	ABD, BCD
Prime	A, B, C, D
Non-prime	E, F
Highest NF	1NF

Q3 R(A,B,C,D,E,F,G,H,I)

FDs: $AB \rightarrow C$ | $BD \rightarrow EF$ | $AD \rightarrow GH$ | $A \rightarrow I$

A, B, D never appear on a RHS $\rightarrow$ compulsory. $(ABD)^+$ reaches all 9 attributes; no 2-subset suffices.

$AB \subset$ key ABD and $AB \rightarrow C$ gives non-prime C $\rightarrow$ partial dependency $\rightarrow$ 2NF fails.

Candidate Key	ABD
Prime	A, B, D
Non-prime	C, E, F, G, H, I
Highest NF	1NF

Q4 R(A,B,C,D,E)

FDs: $AB \rightarrow CD$ | $D \rightarrow A$ | $BC \rightarrow DE$

B is compulsory (never on RHS) but insufficient alone.

$(AB)^+$, $(BC)^+$, $(BD)^+$ all reach $\{A, B, C, D, E\} \rightarrow$ three minimal keys.

Only E is non-prime and it's never derived from a lone proper subset $\rightarrow$ 2NF holds.

Every FD has a superkey LHS or prime RHS ($D \rightarrow A$, A is prime) $\rightarrow$ 3NF holds.

For BCNF, $D \rightarrow A$ needs D to be a superkey — it isn't, so BCNF fails even though A is prime.

Candidate Keys	AB, BC, BD
Prime	A, B, C, D
Non-prime	E
Highest NF	3NF

Q5 R(A,B,C,D,E)

FDs: $BC \rightarrow ADE$ | $D \rightarrow B$

C is compulsory (never on RHS) but insufficient alone.

$(BC)^+ = \text{all attrs directly}$; $(CD)^+ = \text{all attrs via } D \rightarrow B \text{ then } BC \rightarrow ADE \rightarrow \text{two minimal keys}$.

Non-prime A, E never derived from a lone proper subset $\rightarrow$ 2NF holds.

$D \rightarrow B$: D isn't a superkey, but B is prime $\rightarrow$ 3NF exception applies $\rightarrow$ 3NF holds; BCNF fails since D isn't a superkey.

Candidate Keys	BC, CD
Prime	B, C, D
Non-prime	A, E
Highest NF	3NF

Q6 R(A,B,C)

FDs: $A \rightarrow B$ | $B \rightarrow C$ | $C \rightarrow A$

Each single attribute's closure reaches $\{A,B,C\} \rightarrow A, B, C$ are each independent minimal keys (cyclic FDs).

Every attribute is prime $\rightarrow$ no non-prime attributes $\rightarrow$ 2NF, 3NF automatic.

Every FD's LHS (A, B, C individually) is itself a key $\rightarrow$ BCNF fully satisfied.

Candidate Keys	$\{A\}, \{B\}, \{C\}$
Prime	A, B, C (all)
Non-prime	None
Highest NF	BCNF

05 PRACTICAL SET B — NF, KEY & DECOMPOSITION

Note: Q1's FD $H \rightarrow J$ introduces J, so R is treated as 10 attributes (A...J), a minor typo in the original listing.

Q1 R(A,B,C,D,E,F,G,H,I,J)

FDs: $AB \rightarrow C$ | $AD \rightarrow GH$ | $BD \rightarrow EF$ | $A \rightarrow I$ | $H \rightarrow J$

A, B, D never appear on a RHS $\rightarrow$ compulsory. $(ABD)^+$ chains through every FD to reach all 10 attributes; no pair suffices.

$AB \subset \text{key } ABD$ and $AB \rightarrow C$ gives non-prime C $\rightarrow$ partial dependency $\rightarrow$ 2NF fails.

Candidate Key	ABD
Prime	A, B, D
Non-prime	C,E,F,G,H,I,J
Highest NF	1NF

Decomposition (3NF Synthesis)

Relation	Derived From
R1(A,B,C)	$AB \rightarrow C$

Relation	Derived From
R2(A,D,G,H)	AD → GH
R3(B,D,E,F)	BD → EF
R4(A,I)	A → I
R5(H,J)	H → J
R6(A,B,D)	candidate-key relation (added for lossless join)

Total new tables = 6 (R1–R6) — each in 3NF/BCNF; decomposition is lossless and dependency-preserving.

Q2 R(A,B,C,D,E)

FDs: $A \rightarrow B$ | $B \rightarrow E$ | $C \rightarrow D$

A, C never appear on a RHS → compulsory. $(AC)^+$ = all attrs; neither alone suffices → unique key AC.

$A \subset \text{key } AC$ with $A \rightarrow B$ (non-prime B), and $C \rightarrow D$ (non-prime D) → both partial dependencies → 2NF fails.

Candidate Key	AC
Prime	A, C
Non-prime	B, D, E
Highest NF	1NF

Decomposition (one relation per FD)

Relation	Derived From
R1(A,B)	$A \rightarrow B$
R2(B,E)	$B \rightarrow E$
R3(C,D)	$C \rightarrow D$
R4(A,C)	candidate-key relation (added for lossless join)

Total new tables = 4 (R1–R4), all in BCNF; their join reconstructs R losslessly.

06 PRACTICAL SET C — PRIME / NON-PRIME ATTRIBUTES

Q P(F1,F2,F3,F4,F5)

FDs: $F1 \rightarrow F3$ | $F2 \rightarrow F4$ | $(F1, F2) \rightarrow F5$

F1, F2 never appear on a RHS → compulsory. $(F1F2)^+$ = all 5 attributes; neither alone suffices → unique key F1F2.

$F1 \rightarrow F3$ and $F2 \rightarrow F4$ each depend on only part of key F1F2 → partial dependencies → 2NF fails.

$(F1, F2) \rightarrow F5$ is a full dependency on the whole key — not itself a violation.

Candidate Key	F1F2
Prime	F1, F2
Non-prime	F3, F4, F5
Highest NF	1NF

07 RESULT

Candidate keys, prime/non-prime attributes and the highest satisfied normal form were determined for all nine relations via attribute-closure analysis. Relations violating 2NF were decomposed using FD-based synthesis into lossless-join, dependency-preserving schemas, with new-table counts reported above.

08 LEARNING OUTCOMES

- Computing attribute-set closure under a given FD set.
- Systematic identification of all candidate keys, including multi-key cases.
- Classifying attributes as prime / non-prime.
- Testing a relation against 1NF–BCNF conditions.
- Distinguishing partial dependency (2NF) from transitive dependency (3NF).
- Applying FD-based synthesis for lossless, dependency-preserving decomposition.
- Understanding why BCNF is stricter than 3NF.